

Statistics of Financial Markets

Chapter 12: Scoring models



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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AR**: Accuracy Ratio (of the CAP curve)
- **AUC**: Area Under the (ROC) Curve
- **BIS**: Bank for International Settlements
- **BS**: Brier Score
- **CAP**: Cumulative Accuracy Profile
- **CDS**: Credit Default Swap
- **CI**: Confidence Interval
- **DD**: Distance to Default
- **DM**: Deutsche Mark (the German mark)
- **EAD**: Exposure At Default
- **EBIT**: Earnings Before Interest and Taxes
- **EU**: European Union
- **FICO**: Fair Isaac Corporation
- **FN**: False Negative
- **FPR**: False Positive Rate
- **GBM**: Geometric Brownian Motion
- **HL**: Hosmer–Lemeshow (test)
- **IRB**: Internal Ratings-Based (approach)

Acronyms used in this material (2/2)

- **KS**: Kolmogorov–Smirnov (statistic)
- **LDA**: Linear Discriminant Analysis
- **LGD**: Loss Given Default
- **LR**: Likelihood Ratio (test)
- **MVA**: Applied Multivariate Statistical Analysis (Härdle and Simar)
- **NYU**: New York University
- **PD**: Probability of Default
- **PDO**: Points to Double the Odds
- **ROC**: Receiver Operating Characteristic (curve)
- **RWA**: Risk-Weighted Assets
- **SFE**: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)
- **TA**: Total Assets
- **TN**: True Negative
- **TP**: True Positive
- **TPR**: True Positive Rate
- **UCI**: University of California, Irvine (Machine Learning Repository)
- **US**: United States

Today's Question and Route

- **Question**: how does a bank turn the data of an applicant into a probability of default, and how do we know that the number is right?
  - Chapters 2–10 measured the risk of prices; this chapter measures the risk that a borrower does not pay back
- **Route** of the chapter
  - credit scores, PD, LGD, EAD and the Basel capital; default data and class imbalance
  - Weight of Evidence and Information Value; logistic regression and Fisher's discriminant analysis
  - the scorecard and its validation: confusion matrix, ROC, AUC, Gini, CAP, KS, Brier score, calibration
  - reject inference and fairness; the Altman Z-score and the Merton distance to default

Learning Outcomes

- Define PD, LGD, EAD and the expected loss, and explain what the Basel IRB capital adds to the expected loss
- Compute the Weight of Evidence and the Information Value of a variable from a table of counts
- Estimate a logistic regression and interpret its coefficients as changes in log-odds and as odds ratios
- Derive Fisher's linear discriminant and compare it with the logit
- Scale a scorecard and validate it out of sample with the confusion matrix, AUC, Gini, KS, the Brier score and a calibration plot

Reading and Tools

- Textbook: Franke, Härdle & Hafner (2019), Ch. 21 (probability of default) and Ch. 22 (credit risk management)
  - discriminant analysis: Härdle & Simar (2019), Ch. 14
  - credit scoring: Thomas, Crook & Edelman (2017); scorecards: Siddiqi (2006)
- Python Quantlets of this chapter: Quantlets/Ch_12
  - the conditional PD of the one-factor model is ported from the SFE Quantlet SFEdefaproba
- Lecture notebook: open in Google Colab
- Video course: MVA Multivariate Statistical Analysis

The Lending Decision

- An applicant asks for a loan; the bank must decide **accept** or **reject**, and at which interest rate
  - **default**: the borrower does not pay as agreed; Basel: the bank judges the borrower unlikely to pay, or the borrower is more than 90 days past due on a material obligation Basel Committee (2006), ¶452
- **Credit score**: a number computed from the data of an applicant that orders applicants by their risk of default
  - **PD** (probability of default): the probability of default within a horizon, usually one year
  - a good score both **ranks** (discrimination) and gives the right **level** of PD (calibration)
- Kinds of scores
  - application scores (new clients) and behavioural scores (existing clients, from their payment history)
  - consumer scores (statistical models on many small loans) and corporate scores (accounting ratios, market prices)

From Durand to FICO

- 1941: Durand (1941) separates good and bad instalment loans of US lenders with Fisher's discriminant analysis
  - the first statistical credit-scoring study
- 1956: Bill Fair and Earl Isaac found Fair, Isaac and Company (today FICO) myFICO, history of the FICO score
  - 1989: the FICO score, computed from credit-bureau reports, between 300 and 850 points
- Statistical scoring replaced the judgement of a loan officer with a rule that is the same for everybody Hand & Henley (1997)
  - cheaper, faster and testable; the question whether the rule is fair remains



FICO headquarters, San Jose, California

Photo: Coolcaesar (2020); CC BY-SA 4.0; Wikimedia Commons

PD, LGD, EAD and the Expected Loss

- Three risk parameters of a loan (Basel II, Basel Committee (2006))
  - **PD**: probability of default within one year
  - **LGD** (loss given default): the share of the exposure lost if default occurs, after recoveries
  - **EAD** (exposure at default): the amount owed at the moment of default
- **Expected loss**: $\mathrm{EL} = \mathrm{PD} \times \mathrm{LGD} \times \mathrm{EAD}$ (with independent PD and LGD)
  - **Example**: a loan of 100 000 RON with PD $= 2\%$, LGD $= 45\%$: EL $= 0.02 \times 0.45 \times 100\,000 = 900$ RON
  - EL is a cost of doing business: it is priced into the interest rate and covered by provisions
- Capital covers the **unexpected loss**: losses above EL in a bad year

The Basel IRB Capital Formula

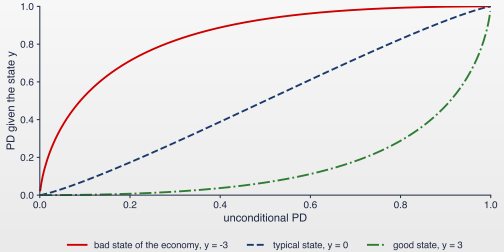
- **IRB** (internal ratings-based) approach: the bank estimates PD (and possibly LGD, EAD); the regulator gives the formula Basel Committee (2006), ¶328–330
  - $K = \mathrm{LGD}\left[\Phi\left(\frac{\Phi^{-1}(\mathrm{PD}) + \sqrt{R}\,\Phi^{-1}(0.999)}{\sqrt{1-R}}\right) - \mathrm{PD}\right]$; RWA $= 12.5\,K\,\mathrm{EAD}$
  - K: the capital per unit of EAD; Φ, Φ^{-1}: the $N(0,1)$ distribution function and its inverse; 0.999: a bad year occurs once in 1000 years
  - R: the correlation of the borrowers with one common factor; RWA: risk-weighted assets, of which the bank holds at least 8%
- Logic Gordy (2003): the term in $\Phi(\cdot)$ is the PD in a year whose common factor is exceeded only with probability 0.1%
  - so K is the credit VaR 0.1% of the loss minus EL: the unexpected loss (Chapter 10)



The BIS tower in Basel

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PD in Good and Bad Years: the One-Factor Model



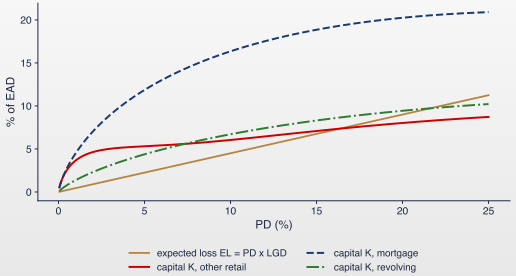
- Borrower i defaults if $\sqrt{\rho}\,Y + \sqrt{1-\rho}\,\varepsilon_i < \Phi^{-1}(\mathrm{PD})$; Y: the state of the economy; given $Y = y$: $\mathrm{PD}(y) = \Phi\big((\Phi^{-1}(\mathrm{PD}) - \sqrt{\rho}\,y)/\sqrt{1-\rho}\big)$, $\rho = 0.2$
- ε_i: the borrower's own shock; Y and ε_i independent $N(0,1)$; ρ: the weight of the common factor
- Interpretation: a loan with PD $= 2\%$ defaults with probability 21.3% in a bad year ($y = -3$), 1.1% in a typical year and almost never in a good year; IRB uses $y = \Phi^{-1}(0.001)$

SFM_ch12_validation_irb

Worked Example: IRB Capital of a Retail Loan

- Other retail exposures: $R = 0.03\,w + 0.16\,(1 - w)$, $w = (1 - e^{-35\,\mathrm{PD}})/(1 - e^{-35})$ Basel Committee (2006), ¶330
  - $w \in [0, 1]$ grows with PD, so R falls from 0.16 (very safe loans) to 0.03 (very risky loans)
  - PD = 2%: $R = 0.095$; $\Phi^{-1}(0.02) = -2.054$, $\Phi^{-1}(0.999) = 3.090$
- **Step by step**
  - argument: $(-2.054 + \sqrt{0.095} \times 3.090)/\sqrt{1 - 0.095} = -1.160$; $\Phi(-1.160) = 12.31\%$: the PD in a bad year
  - $K = 0.45 \times (12.31\% - 2\%) = 4.64\%$ of EAD: 4 639 RON for a loan of 100 000 RON
  - RWA $= 12.5 \times 4\,639 = 57\,986$ RON: a risk weight of 58%
- Interpretation: EL (900 RON) is priced in; the capital (4 639 RON) absorbs a year in which defaults reach 12.31% instead of 2%

Expected Loss and IRB Capital as Functions of PD



- LGD $= 45\%$; capital K for other retail (R from 0.16 to 0.03), residential mortgages ($R = 0.15$) and revolving retail exposures ($R = 0.04$)
- Interpretation: EL grows linearly with PD; K grows fast at low PD and then flattens, because a very risky loan is mostly an expected (priced) loss

SFM_ch12_validation_irb

The South German Credit Data

- 1 000 consumer credits of a large regional bank in southern Germany, 1973–1975 South German Credit (2020); Grömping (2019)
  - outcome: the contract was complied with (good) or not (bad, **default** $= 1$): 300 bad and 700 good credits
  - 20 variables: checking account, duration, credit history, purpose, amount (DM, Deutsche Mark), savings, employment, age and others
- A **stratified sample**: bad credits are heavily oversampled; the bad rate of the bank was about 5% Grömping (2019)
  - all borrowers had passed the bank's checks: rejected applicants are missing (reject inference, Section 8)
  - the cost of accepting a bad credit is set to five times the cost of rejecting a good one
- Licence CC BY 4.0; saved once in the course repository (data/credit)

Coding Errors in the German Credit Data

- The data were used for decades in the UCI version "Statlog (German Credit)", with code labels that were wrong Grömping (2019)
  - in that version, "no checking account" was the safest group: 394 credits, 11.7% bad
  - in the corrected data, those 394 credits have a balance of at least 200 DM or a salary account; "no checking account" is the riskiest group (49.3% bad)
- The numbers were right, the labels were wrong: every model was fine, every interpretation of this variable was reversed
  - checks: compare the frequency tables with the original source (here the book of Fahrmeir and Hamerle), ask whether the signs make economic sense
- We use the corrected South German Credit data throughout

Bad Rate by Checking Account



- Share of bad credits in each group; dashed: the sample average of 30%
- Interpretation: the risk falls from 49.3% (no account) to 11.7% (balance of at least 200 DM): one variable already separates the applicants well

SFM_ch12_woe_iv

Class Imbalance and Oversampling

- Defaults are rare: a few percent of the loans of a bank; a sample with few bads tells little about them
  - banks therefore oversample bads (here 30% instead of about 5%)
- **Accuracy paradox**: the rule "everybody is good" is right for 70% of this sample and for 77.9% of the Taiwan data
  - so accuracy alone is a poor measure; we need measures that look at the bads and the goods separately (Section 7)
- **Taiwan credit card data** Default of Credit Card Clients (2009); Yeh & Lien (2009): 30 000 card holders, default on the October 2005 payment, 22.1% defaults
  - inputs: credit limit, age, the repayment status of April–September 2005, bills and payments; we use them for calibration and fairness

Training and Test Samples

- **Stratified split**: 700 credits for estimation (training), 300 for testing, with 30% bads in both
  - every choice (bins, WoE, variables, coefficients) is made on the training sample only
  - the test sample is used once, at the end: it plays the role of the future applicants
- **Leakage**: information from the test sample that enters the model, e.g. bins chosen on all data
  - it makes the test result too optimistic; with 300 test credits one split is also noisy, so we add cross-validation (Section 7)

Weight of Evidence

- Split a variable into bins $i = 1, \dots, m$; g_i, b_i: goods and bads in bin i; G, B: their totals
  - $\mathrm{WoE}_i = \ln \dfrac{g_i / G}{b_i / B}$: the log of the share of goods over the share of bads in the bin
- Meaning: $\mathrm{WoE}_i = \ln \dfrac{g_i}{b_i} - \ln \dfrac{G}{B}$: the log-odds of being good in the bin minus the log-odds in the whole sample
  - $\mathrm{WoE} > 0$: safer than average; $\mathrm{WoE} < 0$: riskier; $\mathrm{WoE} = 0$: no information
  - a bin without goods or without bads would give $\pm\infty$: add 0.5 to both counts
- **Information Value**: $\mathrm{IV} = \sum_i \left(\dfrac{g_i}{G} - \dfrac{b_i}{B} \right) \mathrm{WoE}_i \geq 0$
  - a symmetric Kullback–Leibler divergence between the distributions of goods and bads across the bins
  - rule of thumb Siddiqi (2006): below 0.02 useless, 0.02–0.1 weak, 0.1–0.3 medium, above 0.3 strong

Worked Example: WoE and IV of the Checking Account

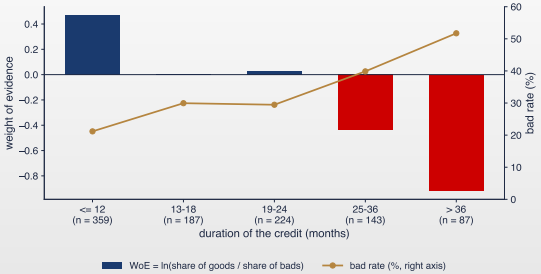
Checking account	goods	bads	g_i/G	b_i/B	WoE	IV term
no checking account	139	135	0.199	0.450	-0.818	0.206
balance $<$ 0 DM	164	105	0.234	0.350	-0.401	0.046
0 to 200 DM	49	14	0.070	0.047	0.405	0.009
$\geq$ 200 DM or salary	348	46	0.497	0.153	1.176	0.404
total	700	300	1	1		0.666

- **Step by step**, last row: $g/G = 348/700 = 0.497$, $b/B = 46/300 = 0.153$
  - WoE $= \ln(0.497/0.153) = 1.176$; IV term $= (0.497 - 0.153) \times 1.176 = 0.404$
- Interpretation: IV $= 0.666 > 0.3$: a strong variable; the WoE falls monotonically from the safest to the riskiest group

Binning: Practical Rules

- Why bins: they capture non-linear effects, limit the influence of outliers and give every applicant a small table of points
  - numeric variables: a few intervals (here duration ≤ 12, 13–18, 19–24, 25–36, > 36 months); categorical: the categories
- Rules Siddiqi (2006); Thomas, Crook & Edelman (2017)
  - at least about 5% of the sample in every bin; merge small categories
  - WoE should change in a way that makes economic sense (often monotonically)
  - bins and WoE are estimated on the training sample and applied unchanged to new applicants
- A WoE variable enters the logit with one coefficient: the scorecard stays additive and easy to read

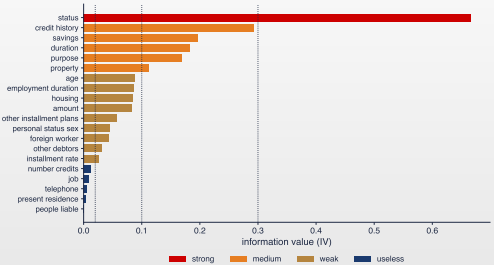
WoE of the Credit Duration



- Full sample; bars: WoE of each duration bin; line: bad rate (right axis); IV = 0.182 (medium)
- Interpretation: credits up to one year are safer (WoE = 0.47, 21% bad); above three years the bad rate reaches 52% (WoE = −0.92); 13–24 months are average

SFM_ch12_woe_iv

Information Value of the 20 Variables



- Full sample; dotted: the thresholds 0.02, 0.1 and 0.3; on the training sample, 6 variables have IV ≥ 0.1 and enter the scorecard
- Interpretation: the checking account (IV $= 0.666$) and the credit history (0.293) dominate; telephone, residence and dependants carry no information
- IV looks at one variable at a time: two strong but correlated variables do not add their information

SFM_ch12_woe_iv

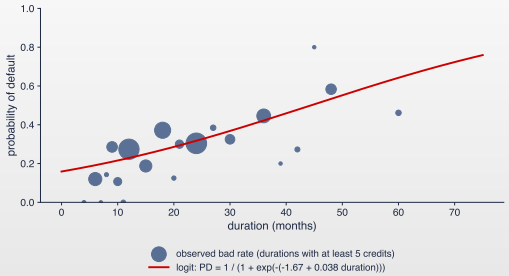
Odds and Log-Odds

- A linear model $P(Y = 1) = \beta_0 + x^\top \beta$ can give probabilities below 0 or above 1
  - we model a transformation of p that can take any real value
- **Odds** of default: $\dfrac{p}{1-p} \in (0, \infty)$; **log-odds** (logit): $\ln \dfrac{p}{1-p} \in (-\infty, \infty)$
  - $p = 0.20$: odds $= 0.2/0.8 = 0.25$ (one default for four repaid loans); log-odds $= \ln 0.25 = -1.386$
  - $p = 0.5$: odds 1, log-odds 0; the logit is symmetric: $\mathrm{logit}(1-p) = -\mathrm{logit}(p)$
- Bankers often speak of the **good:bad odds** $(1-p)/p$: here 4:1

The Logit Model

- $\ln \dfrac{p_i}{1-p_i} = \beta_0 + x_i^\top \beta = \eta_i, \quad p_i = P(Y_i = 1 \mid x_i) = \dfrac{1}{1+e^{-\eta_i}}$ Cox (1958)
  - $x_i = (x_{i1}, \ldots, x_{ik})^\top$: vector of the k characteristics of applicant i; $\beta = (\beta_1, \ldots, \beta_k)^\top$: vector of their coefficients, the same for all applicants
  - β_0: the constant (intercept); $\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}$: the linear predictor, the log-odds of applicant i
- **Maximum likelihood** (Chapter 6): $\ell(\beta) = \sum_i [y_i \ln p_i + (1-y_i)\ln(1-p_i)]$
  - first-order condition $\sum_i (y_i - p_i)x_i = 0$: no closed form, solved by Newton–Raphson
  - standard errors from the inverse of $\sum_i p_i(1-p_i)x_i x_i^\top$; Wald test $z = \hat{\beta}_j / \mathrm{se}(\hat{\beta}_j)$
- The logit is also used for corporate bankruptcy, starting with Ohlson (1980); the probit uses $\Phi(\eta)$ instead of the logistic function
  - logit and probit give almost the same PDs; the logit has the odds interpretation

A Logistic Curve: PD and Duration



- Logit with duration only: $\hat{\beta}_0 = -1.666$, $\hat{\beta}_1 = 0.0375$ per month (se 0.0057); dots: observed bad rates
- Interpretation: each extra year multiplies the odds of default by $e^{12 \times 0.0375} = 1.57$; the PD rises from 22.9% at 12 months to 53.4% at 48 months

SFM_ch12_logit_lda

Interpreting the Coefficients

- β_j: the change in the **log-odds** when x_j grows by one unit, the other variables fixed
  - e^{β_j}: the **odds ratio**: the odds are multiplied by e^{β_j}; $100(e^{\beta_j} - 1)\%$ is the change in the odds
  - a 95% confidence interval: $\exp(\hat{\beta}_j \pm 1.96\,\mathrm{se})$
- The effect on the **probability** depends on where the applicant is
  - $\partial p / \partial x_j = p(1-p)\beta_j$: largest at $p = 0.5$, small for very safe or very risky applicants
- A dummy variable (0/1): e^{β_j} compares the odds of the group with those of the reference group
  - the interpretation needs a model that contains the relevant variables: an omitted variable changes the coefficients

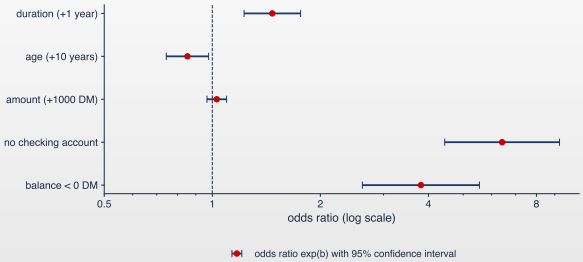
A Small Logit on the Full Sample

Variable	$\hat{\beta}$	se	z	p	odds ratio (95% CI)
constant	−2.101	0.311	−6.75	< 0.001	
duration (years)	0.385	0.092	4.17	< 0.001	1.47 (1.23–1.76)
age (decades)	−0.159	0.069	−2.30	0.022	0.85 (0.74–0.98)
amount (1000 DM)	0.029	0.032	0.90	0.368	1.03 (0.97–1.10)
no checking account	1.859	0.188	9.89	< 0.001	6.42 (4.44–9.28)
balance $<$ 0 DM	1.338	0.192	6.99	< 0.001	3.81 (2.62–5.55)

- $n = 1\,000$; log-likelihood -526.1 against -610.9 with the constant only: LR $= 169.6$ on 5 degrees of freedom
- Reference group of the account dummies: a non-negative balance (0–200 DM, or at least 200 DM or a salary account)

SFM_ch12_logit_lda

Odds Ratios with Confidence Intervals



- Each dot: $e^{\hat{\beta}}$; bars: 95% intervals; log scale; vertical line: no effect (1)
- Interpretation: without a checking account the odds of default are 6.42 times those of the reference group; a balance below zero multiplies them by 3.81
- One more year of duration raises the odds by 47%; ten more years of age lower them by about 15%
- The amount is not significant once the duration is in the model (the two are correlated)

SFM_ch12_logit_lda

Worked Example: the PD of an Applicant

- Applicant: 36 months, 30 years old, 4000 DM, no checking account; coefficients of the previous table, rounded
  - $\eta = -2.101 + 0.385 \times 3 + (-0.159) \times 3 + 0.029 \times 4 + 1.859$
  - $\eta = -2.101 + 1.155 + (-0.477) + 0.116 + 1.859 = 0.552$
- PD $= 1/(1 + e^{-0.552}) = 63.5\%$; odds $e^{0.552} = 1.74$
  - the same applicant with a non-negative balance: $\eta = -1.307$, PD $= 21.3\%$
  - one more year of duration: $\Delta p \approx p(1-p)\beta \approx 8.9$ percentage points
- Caution: this PD refers to a sample with 30% bads, not to the bank's population (prior correction, two slides ahead)

The WoE Logit Scorecard

WoE variable	IV (training)	$\hat{\beta}$	se
constant		−0.850	0.095
checking account	0.591	−0.829	0.127
credit history	0.215	−0.675	0.206
purpose	0.205	−0.937	0.217
savings	0.166	−0.726	0.246
duration	0.159	−1.082	0.239
years with the employer	0.122	−0.873	0.265

- Training sample (700 credits); each variable replaced by the WoE of its bin; logit for default
  - all coefficients are negative: a higher WoE (safer bin) lowers the log-odds of default
  - a coefficient of -1 would mean that the variable adds exactly its own WoE; values between -0.7 and -1.1 reflect the overlap between variables
- Test sample: AUC $= 0.801$ (Section 7); in-sample log-likelihood -347.2

SFM_ch12_scorecard

Correcting for Oversampling

- If bads and goods are sampled with different rates, only the constant of the logit changes
  - $\ln \text{odds}_{\textbf{pop}} = \ln \text{odds}_{\textbf{sample}} + \ln \dfrac{\pi/(1-\pi)}{\pi_s/(1-\pi_s)}$; π: population bad rate, π_s: sample rate
  - the slopes are unchanged: the ranking of the applicants does not depend on the sampling
- **Step by step**: $\pi_s = 0.30$, $\pi = 0.05$: shift $= \ln(0.05/0.95) - \ln(0.30/0.70) = -2.944 - (-0.847) = -2.097$
  - a sample PD of 30% becomes 5%; a sample PD of 50% becomes 10.9%
  - the applicant of the worked example: $0.552 - 2.097 = -1.545$, PD $= 17.6\%$ in the bank's population instead of 63.5%
- Interpretation: AUC and Gini are unchanged by the correction; EL and capital are not

Fisher's Idea (1936)

- Fisher (1936): find the linear combination $w^\top x$ that separates two groups best
  - he used measurements of iris flowers; Durand applied it to loans five years later
- Fisher criterion: $J(w) = \dfrac{[w^\top (m_1 - m_0)]^2}{w^\top S_W w}$
  - m_0, m_1: mean vectors of goods and bads; S_W: the pooled within-group covariance matrix
  - maximum: $w \propto S_W^{-1}(m_1 - m_0)$ Härdle & Simar (2019), Ch. 14
- No distribution is needed for this criterion: it is a projection that maximises the distance between the groups relative to their spread



R.A. Fisher, 1913

Photo: unknown author (1913); public domain; Wikimedia Commons



Iris versicolor

Photo: Ermell (2020); CC BY-SA 4.0; Wikimedia Commons

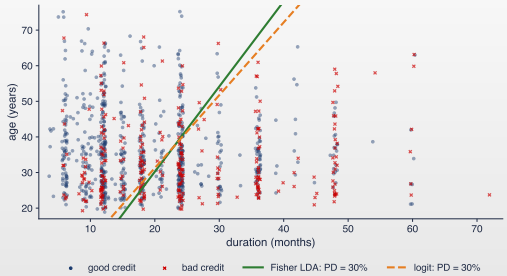
LDA as a Probability Model

- Assume $x \mid Y = k \sim N(m_k, \Sigma)$, the same Σ in both groups, and prior $\pi_1 = P(Y = 1)$
  - Bayes: $\ln \dfrac{P(Y=1 \mid x)}{P(Y=0 \mid x)} = \underbrace{\ln \dfrac{\pi_1}{1-\pi_1} - \tfrac{1}{2}(m_1 + m_0)^\top w}_{c} + w^\top x, \quad w = \Sigma^{-1}(m_1 - m_0)$
  - the log-odds are **linear in** x, exactly as in the logit; the prior enters only the constant
- The difference is in the estimation
  - LDA: plug in the sample means and the pooled covariance (a closed form)
  - logit: maximise the conditional likelihood of Y given x; no assumption on the distribution of x
- Classification rule: assign to "bad" if $c + w^\top x > \ln(C_{FP}/C_{FN})$ (costs, Section 7)

Worked Example: Fisher's Direction with Two Variables

- $x =$ (duration in months, age in years), full sample: 700 goods, 300 bads
  - $m_0 = (19.21, 36.22)$, $m_1 = (24.86, 33.96)$, $m_1 - m_0 = (5.65, -2.26)$
  - $S_W = \begin{pmatrix} 138.8 & -2.46 \\ -2.46 & 127.9 \end{pmatrix}$, $\det S_W = 17756$
- **Step by step**: $S_W^{-1} = \frac{1}{\det S_W} \begin{pmatrix} 127.9 & -(-2.46) \\ -(-2.46) & 138.8 \end{pmatrix}$, $w = S_W^{-1}(m_1 - m_0) = (0.0404, -0.0169)$
  - logit with the same two variables: slopes $(0.0375, -0.0183)$
  - ratio duration/age: LDA -2.39, logit -2.05: one month of duration weighs as much as about two years of age, in the other direction

Fisher's Boundary and the Logit Boundary



- Duration and age of every credit (a small jitter added); lines: the points with PD = 30% (the sample rate) for the LDA and for the logit
- Interpretation: both boundaries are lines with almost the same slope; the two clouds overlap heavily, so two variables are far from enough

SFM_ch12_logit_lda

Logit or LDA?

	Logit	LDA (Fisher)
model	$P(Y \mid x)$ only	$P(x \mid Y)$ Normal, common Σ
estimation	maximum likelihood, iterative	means and covariance, closed form
if x is Normal	consistent, less efficient	more efficient
dummies, WoE, skewed x	fine	ranking fine, PDs may be off
test AUC (WoE inputs)	0.801	0.801

- Test sample: the correlation of the two log-odds scores is 0.999; DeLong test of equal AUCs: $z = 0.18$, p $= 0.86$ DeLong et al. (1988)
- Interpretation: for ranking, the two methods are practically identical; the logit is preferred for PDs because it does not need Normal inputs

SFM_ch12_logit_lda

Scaling a Scorecard

- **Scorecard**: a table that gives points to every attribute; the score of an applicant is the sum of the points Siddiqi (2006)
  - higher score = lower risk; the score is a linear function of the good:bad log-odds
- $\text{Score} = \text{Offset} + \text{Factor} \times \ln \frac{1-p}{p}$, two conventions fix the scale
  - **PDO** (points to double the odds): +PDO points $\Leftrightarrow$ the good:bad odds double: $\text{Factor} = \text{PDO}/\ln 2$
  - a base score at given odds: $\text{Offset} = \text{Base} - \text{Factor} \ln(\text{odds}_0)$
- Here: PDO = 20, 600 points at odds 50:1
  - $\text{Factor} = 20/\ln 2 = 28.85$; $\text{Offset} = 600 - 28.85 \times \ln 50 = 600 - 28.85 \times 3.912 = 487.12$

Worked Example: Scores and Points

- **PD to score**: PD $= 5\%$, odds 19:1: Score $= 487.12 + 28.85 \times \ln 19 = 487.12 + 28.85 \times 2.944 = 572.1$
  - PD $= 30\%$ (odds 7:3): $487.12 + 28.85 \times 0.847 = 511.6$
- **Score to PD**: 560 points: ln odds $= (560 - 487.12)/28.85 = 2.526$, odds 12.5:1, PD $= 7.4\%$
- **Points of an attribute** ($k = 6$ variables; logit for default with constant β_0 and slope β_j)
  - $\text{points}_{ij} = -(\beta_j \, \text{WoE}_{ij} + \beta_0/k)\,\text{Factor} + \text{Offset}/k$; the minus sign turns default log-odds into good:bad log-odds
  - checking account (training WoE): no account 66 points, balance $<$ 0 77, 0–200 DM 97, at least 200 DM or salary 110
- Prior correction in points: for a population with 5% bads the default log-odds fall by 2.097, so the good:bad log-odds rise by 2.097 and every score rises by $28.85 \times 2.097 = 60.5$ points

SFM_ch12_scorecard

Scores of the Test Sample



- 300 test credits; scores from 422 to 607 points; mean: goods 529, bads 492; dashed: the cut-off at PD $= 1/6$ (Section 7), 534 points
- Interpretation: the two distributions are shifted but overlap: every cut-off rejects some goods and accepts some bads; the validation measures quantify this overlap

SFM_ch12_scorecard

The Confusion Matrix

	predicted bad (reject)	predicted good (accept)
actual bad	TP	FN
actual good	FP	TN

- A cut-off c: reject if PD $> c$
- TP (true positive): a bad rejected; FN (false negative): a bad accepted
- FP (false positive): a good rejected; TN (true negative): a good accepted

- Rates
  - **TPR** (true positive rate, sensitivity) $= \mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$: share of bads caught
  - **FPR** (false positive rate) $= \mathrm{FP}/(\mathrm{FP}+\mathrm{TN}) = 1-$ specificity: share of goods rejected
  - precision $= \mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$; accuracy $= (\mathrm{TP}+\mathrm{TN})/n$
- Every rate depends on c; the measures that follow summarise all cut-offs at once

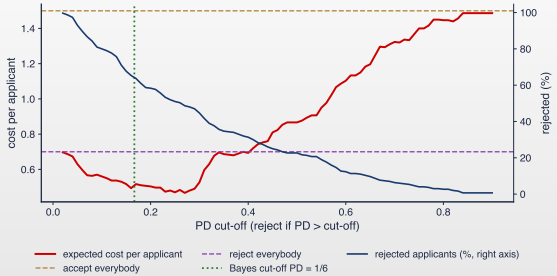
Two Cut-Offs on the Test Sample

cut-off	TP	FN	FP	TN	accuracy	TPR	FPR	cost
PD > 0.5	43	47	25	185	76.0%	47.8%	11.9%	0.867
PD $> 1/6$	82	8	112	98	60.0%	91.1%	53.3%	0.507

- Cost per applicant $= (5\,\mathrm{FN} + 1\,\mathrm{FP})/n$: accepting a bad costs five times as much as rejecting a good Grömping (2019)
  - the Bayes cut-off minimises the expected cost: reject if $p\,C_{FN} > (1-p)\,C_{FP}$, i.e. $p > C_{FP}/(C_{FP}+C_{FN}) = 1/6$
- Interpretation: the cut-off of 1/6 has a lower accuracy (60.0% against 76.0%) but a much lower cost (0.507 against 0.867): it catches 91.1% of the bads

SFM_ch12_validation_irb

Expected Cost and the Choice of the Cut-Off



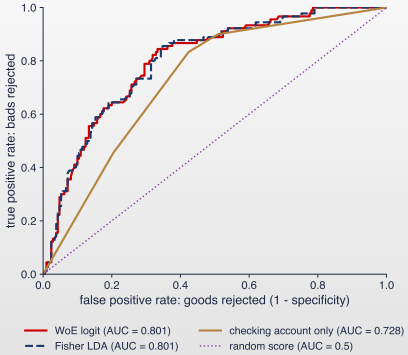
- Test sample; red: cost per applicant; dashed: accept everybody (1.50) or reject everybody (0.70); blue: share rejected (right axis)
- Interpretation: the cost is flat between PD 0.1 and 0.3; the minimum on these 300 credits is at 0.27, close to the Bayes cut-off of 1/6, which is chosen before seeing the test data

SFM_ch12_validation_irb

ROC Curve and AUC

- **ROC** (receiver operating characteristic): the points (FPR(c), TPR(c)) for all cut-offs c
  - from (0, 0) (reject nobody) to (1, 1) (reject everybody); the diagonal is a random score
- **AUC** (area under the curve) $= P(s_{\text{bad}} > s_{\text{good}}) + \frac{1}{2} P(s_{\text{bad}} = s_{\text{good}})$ Hanley & McNeil (1982)
  - s: the risk score (here the PD); compute it by counting pairs: the share of (bad, good) pairs ordered correctly
  - the Mann–Whitney statistic divided by $n_1 n_0$; 0.5: no discrimination, 1: perfect
- **Gini** $= 2\,\mathrm{AUC} - 1$; AUC does not depend on the bad rate or on a monotone rescaling of the score
  - standard error and tests of AUC: DeLong et al. (1988)

ROC Curves on the Test Sample



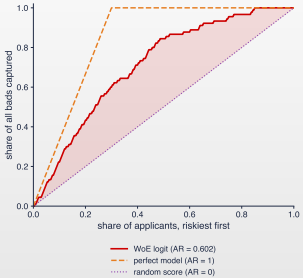
- 300 test credits, 18 900 (bad, good) pairs
- WoE logit: AUC $= 0.801$, 95% CI $[0.749, 0.854]$ (DeLong, se 0.027); Gini $= 0.603$
- Checking account alone: AUC $= 0.728$; DeLong test against the scorecard: $z = 2.86$, $p = 0.004$
- Interpretation: a random bad has a higher PD than a random good in about 80% of the pairs; the other five variables add a significant improvement over the checking account

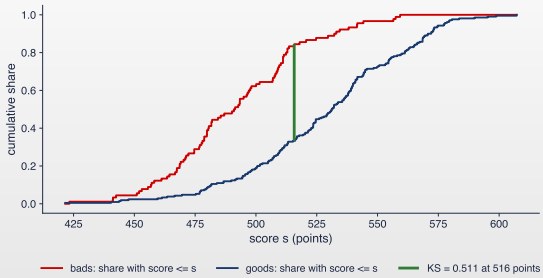
SFM_ch12_validation_irb

CAP Curve, Accuracy Ratio and KS

- **CAP** (cumulative accuracy profile): sort the applicants from the riskiest; plot the share of all bads among the first x%
  - perfect model: all bads first, the curve reaches 1 at $x =$ bad rate; random: the diagonal
  - **AR** (accuracy ratio) $= \dfrac{\text{area between model CAP and diagonal}}{\text{area between perfect CAP and diagonal}}$; for a continuous score AR $=$ Gini
- **KS** (Kolmogorov–Smirnov): $\max_s |F_{\text{bad}}(s) - F_{\text{good}}(s)|$, the largest gap between the score distributions of bads and goods
  - F_{bad}, F_{good}: the empirical distribution functions of the scores in the two groups; KS $\in [0, 1]$
  - read on the ROC curve: KS $= \max_c (\mathrm{TPR} - \mathrm{FPR})$; the score where it occurs is a natural first cut-off
- All three measure **discrimination** (ranking), not calibration

CAP and KS on the Test Sample





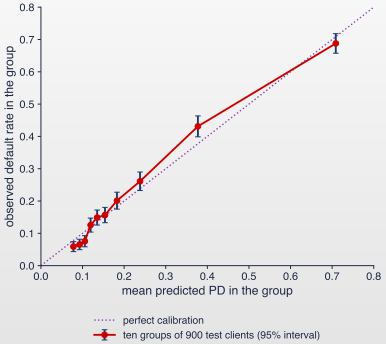
- AR = 0.602 (Gini = 0.603: the same up to rounding); KS = 0.511 at 516 points
- Interpretation: rejecting the riskiest 30% catches 60% of the bads (54 of 90); at 516 points the scorecard separates bads and goods best

SFM_ch12_validation_irb

Calibration and the Brier Score

- **Calibration**: among applicants with PD $\approx p$, a share of about p defaults
  - reliability diagram: sort by PD, form groups (e.g. ten), plot the mean PD against the observed default rate
  - Hosmer–Lemeshow Hosmer & Lemeshow (1980): $\sum_g \frac{(O_g - n_g \bar{p}_g)^2}{n_g \bar{p}_g (1 - \bar{p}_g)} \approx \chi^2(G-2)$
  - G groups; n_g: size, O_g: observed defaults, $\bar{p}_g$: mean PD of group g; a large value: poor calibration
- **Brier score** Brier (1950): $\mathrm{BS} = \frac{1}{n}\sum_i (p_i - y_i)^2$, the mean squared error of the PDs; lower is better
  - reference: a constant PD equal to the training bad rate; skill $= 1 - \mathrm{BS}/\mathrm{BS}_{\mathrm{ref}}$
  - scorecard test sample: BS $= 0.162$ against 0.210 (skill 23%); HL with 10 groups $= 6.520$, p $= 0.589$
- Brier rewards both ranking and calibration; AUC sees only the ranking

Calibration on the Taiwan Data



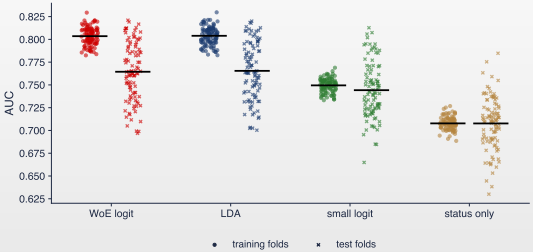
- Logit on 30 000 card holders (70% estimation); test sample of 9 000; ten groups of 900 by PD; bars: 95% intervals
- Test AUC $= 0.777$ (estimation 0.774); Brier $= 0.136$ against 0.172 (skill 21%)
- Hosmer–Lemeshow $= 40.660$, $p < 0.001$: rejected, although the points lie close to the diagonal
- Interpretation: with 9000 clients the test detects small deviations (the safest group: PD 7.8%, observed 5.9%); judge the size of the gaps, not only the p-value

SFM_ch12_validation_irb

Out-of-Sample and Cross-Validation

- **In-sample** performance is optimistic: the model has adapted to the noise of the training data (overfitting)
  - one test sample of 300 credits gives AUC $= 0.801$ with a standard error of 0.027: one split is noisy
- k**-fold cross-validation**: split into $k = 5$ folds; estimate on four, test on the fifth; rotate; repeat 20 times
  - the whole procedure is repeated inside every fold: bins, WoE, IV selection and coefficients
  - report the mean and the spread of the 100 test AUCs, and the gap between training and test AUC
- **Out-of-time** validation: estimate on older applications, test on later ones (two slides ahead)

Training and Test AUC in Repeated Cross-Validation



- 20 repetitions of 5 folds; dots: training folds, crosses: test folds; black lines: means
- WoE logit: training 0.804, test 0.764 (sd 0.032); LDA: 0.804, 0.765; small logit: 0.749, 0.744; checking account: 0.708
- Interpretation: the scorecard loses 0.039 AUC out of sample, the five-variable logit almost nothing; the single split above was lucky (0.801 against 0.764 on average)

SFM_ch12_validation_irb

Out-of-Time Validation and Drift

- A scorecard is used for years on applicants who arrive **after** the estimation period
  - the economy, the products and the applicants change: the relation between x and default drifts
  - out-of-time test: estimate on, say, 2019–2021, test on 2022–2023; compare AUC and calibration across years
- Monitoring: the distribution of the scores and of each variable is compared with the training period
  - a shift with stable ranking calls for recalibration (the constant); a loss of ranking calls for a new model
- Data limitation: neither data set of this chapter has application dates, so an out-of-time test is not possible here; random splits are the most we can do
- Further reading, an out-of-time design for crypto "zombie" assets: Będowska-Sójka, Wójcik & Pele (2026)

Case Study: Benchmarking Credit Scoring Methods

- Lessmann et al. (2015): 41 classifiers, six accuracy measures, eight retail credit data sets (among them the German credit data of UCI)
  - an update of the benchmark of Baesens et al. (2003), ten years later
- Findings
  - several classifiers predict significantly better than logistic regression, the industry standard; heterogeneous ensembles perform best
  - more accurate scorecards can bring sizeable financial returns
  - the common accuracy measures give similar signals; random forests are recommended as the benchmark for any new method
- Beating the logit alone is no longer evidence of progress; on our 1000 credits, logit and LDA differ by less than one standard error of AUC; Chapter 13 adds trees and neural networks

Reject Inference

- The outcome is known only for **accepted** applicants; the model is used on **all** applicants ("through the door")
  - a sample selected by the old score: the riskiest applicants are missing, and so are their outcomes
  - the South German Credit borrowers had all passed the bank's checks Grömping (2019)
- **Reject inference**: methods that try to correct for the missing outcomes Banasik, Crook & Thomas (2003); Thomas, Crook & Edelman (2017)
  - reweighting: give accepted applicants who resemble the rejected ones a larger weight
  - augmentation and parcelling: impute outcomes for the rejected from a model
  - the most reliable source: a small random sample of applicants accepted regardless of the score

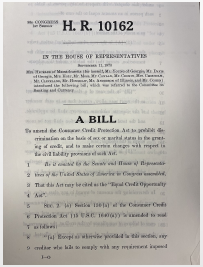
A Reject-Inference Experiment on the Taiwan Data

- An "old score" (late payment in September and the limit) accepts the best 70% of the training clients (15 085); their default rate is 13.2%, that of the rejected 44.9%
  - among the accepted nobody was late in September: the strongest variable cannot be estimated and drops out
- Test sample (all applicants): AUC of the model estimated on the accepted only 0.736, on all applicants 0.777, old score 0.743
  - rejected test clients: observed default rate 46.1%; mean PD of the accepted-only model 29.6%, of the full model 44.1%
- Interpretation: trained on the accepted only, the new model ranks worse than the old score and underestimates the risk of the rejected by about 36%

SFM_ch12_reject_fairness

Fairness in Credit Scoring

- US law forbids discrimination in credit by sex, marital status, race, religion, national origin or age Equal Credit Opportunity Act (FTC)
  - EU: credit scoring of persons is a high-risk use of AI (documentation, human oversight) European Commission, AI Act
- Leaving the protected variable out is not enough: other variables can stand in for it
  - **demographic parity**: equal approval rates across groups
  - **equal opportunity**: equal approval rates of the good payers Hardt, Price & Srebro (2016)
  - **calibration within groups**: the same PD means the same risk in every group
- When default rates differ across groups, these criteria cannot all hold at once: the choice is a matter of policy for banks and regulators



H. R. 10162

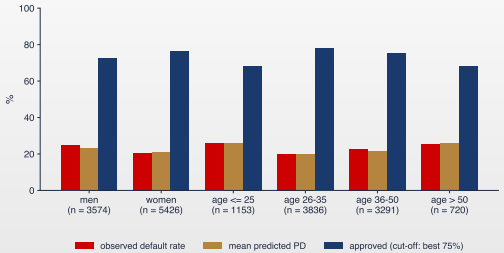
IN THE HOUSE OF REPRESENTATIVES

A BILL

H.R. 10162 (1973), the bill that introduced the Equal Credit Opportunity Act

Photo: H.R. 10162 (1973); public domain; Wikimedia Commons

Approval and Default by Group: Taiwan



- Test sample; sex and age are not inputs of the model; cut-off: approve the 75% with the lowest PD
- Women: default rate 20.5%, approved 76.6%; men: 24.7%, 72.5%; good payers approved: women 85.1%, men 83.2%
- Interpretation: the model is calibrated in every group (mean PD close to the observed rate), so groups with more defaults (men, the youngest and the oldest) are approved less often

SFM_ch12_reject_fairness

Case Study: Machine Learning and Unequal Credit

- Fuster et al. (2022): US mortgages; default predicted with a traditional logit and with machine-learning models
  - question: who gains and who loses when the lender switches to the more accurate model?
- Findings
  - the flexible models predict default better
  - Black and Hispanic borrowers are less likely to gain from the new technology
  - in an equilibrium model, the disparity in interest rates between and within groups increases, mainly because of the greater flexibility
- Lesson: accuracy and fairness must be measured together; a better AUC is not a sufficient argument

The Altman Z-Score (1968)

- Altman (1968): 66 US manufacturing firms, 33 that filed for bankruptcy in 1946–1965 and 33 matched survivors
  - Fisher's discriminant analysis on five accounting ratios
- $Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$
  - X_1 working capital / total assets (TA); X_2 retained earnings / TA; X_3 EBIT (earnings before interest and taxes) / TA
  - X_4 market value of equity / book value of debt; X_5 sales / TA
  - zones: $Z < 1.81$ distress, 1.81–2.99 grey, $Z > 2.99$ safe
- About 95% of the sample correctly classified one year before bankruptcy; the accuracy falls quickly for longer horizons



NYU Stern School of Business, Altman's institution

Photo: Ajay Suresh (2019); CC BY 2.0; Wikimedia Commons

Worked Example: a Z-Score

- A hypothetical firm: $X_1 = 0.15$, $X_2 = 0.20$, $X_3 = 0.08$, $X_4 = 0.90$, $X_5 = 1.10$
  - $Z = 1.2 \times 0.15 + 1.4 \times 0.20 + 3.3 \times 0.08 + 0.6 \times 0.90 + 1.0 \times 1.10$
  - $Z = 0.180 + 0.280 + 0.264 + 0.540 + 1.100 = 2.364$: the grey zone
- Which ratio would move the firm to the safe zone?
  - an EBIT/TA of 0.27 instead of 0.08 adds $3.3 \times 0.19 \approx 0.63$: $Z \approx 2.99$
- Limits
  - the weights come from 66 US manufacturers of 1946–1965; other sectors and countries need re-estimation
  - Z is a discriminant score, not a PD; later models give PDs: logit Ohlson (1980), hazard models Shumway (2001)

Merton's Distance to Default

- Merton (1974): the value V_t of the firm's assets follows a GBM (geometric Brownian motion, Chapter 4) with drift μ and volatility σ_V
  - debt with face value D due at T; default if $V_T < D$; equity is a call option on the assets
- $\ln V_T \sim N\big(\ln V_0 + (\mu - \sigma_V^2/2)T,\ \sigma_V^2 T\big)$, so
  - $\mathrm{DD} = \dfrac{\ln(V_0/D) + (\mu - \sigma_V^2/2)T}{\sigma_V\sqrt{T}}, \quad \mathrm{PD} = \Phi(-\mathrm{DD})$
  - DD: the number of standard deviations between the expected asset value and the debt
- V and σ_V are not observed: they are solved from the equity price and the equity volatility (option pricing)



Robert C. Merton, 2010

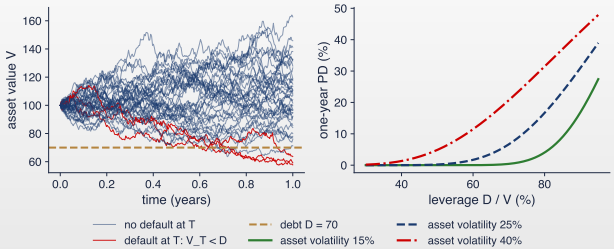
Photo: Massachusetts Institute of Technology (2010); CC BY-SA 4.0; Wikimedia Commons

Worked Example: Distance to Default

- $V_0 = 100$, $D = 70$, $\sigma_V = 25\%$, $\mu = 5\%$, $T = 1$ year
  - $\ln(100/70) = 0.3567$; $\mu - \sigma_V^2/2 = 0.05 - 0.03125 = 0.01875$
  - $\mathrm{DD} = 0.3754/0.25 = 1.502$; $\mathrm{PD} = \Phi(-1.502) = 6.66\%$
- From the market: equity 40, equity volatility 50%, debt 70, risk-free rate 3%
  - solving $E = V\Phi(d_1) - De^{-rT}\Phi(d_2)$ and $\sigma_E E = \Phi(d_1)\sigma_V V$: $V = 107.9$, $\sigma_V = 18.6\%$, $\mathrm{DD} = 2.39$, $\mathrm{PD} = 0.84\%$
  - E, σ_E: value and volatility of equity; r: the risk-free rate; d_1, d_2: the Black–Scholes terms of Chapter 4, with V in place of the share price
- Interpretation: a market-based PD reacts every day to prices; an accounting score changes once a year

SFM_ch12_altman_merton

Merton: Asset Paths and PD



- Left: 40 simulated asset paths ($V_0 = 100$, $\mu = 5\%$, $\sigma_V = 25\%$); red: the paths that end below the debt of 70
- Right: one-year PD against leverage D/V for three asset volatilities
- Interpretation: PD is almost zero at low leverage and rises steeply between 70% and 90%; the higher the volatility, the earlier the rise

SFM_ch12_altman_merton

Case Study: How Good Is the Merton Model?

- Bharath & Shumway (2008): US firms; the Merton DD against a "naive" DD with the same functional form but without solving the model
  - design: hazard models of default and out-of-sample forecasts; comparison with CDS (credit default swap) spreads and bond yields
- Findings
  - the naive DD performs slightly better than the Merton DD out of sample
  - other variables (e.g. past returns) add information: DD is not a sufficient statistic for PD
  - its functional form (leverage scaled by volatility) is what makes it useful
- Lesson: theory gives a good variable, statistics decides how to use it; the same holds for the Z-score

An Open Question

- **Do flexible machine-learning scores improve default prediction enough to justify their cost in transparency and fairness?**
  - today: WoE logit and LDA reach a test AUC of about 0.764; benchmarks find significant gains for ensembles Lessmann et al. (2015)
  - the gains may be unequal across groups Fuster et al. (2022); support vector machines for default: Chen, Härdle & Moro (2011)
- Why it is open: gains depend on the data set, the measure and the population; the outcomes of rejected applicants are never seen
- AI tools can speed up such a study, but its results must still be checked Wang et al. (2023)

Possible Uses of AI Tools

- **Literature**: a list of benchmark studies of credit scoring and of studies on fairness in lending, with their data sets
- **Code**: a first draft of a cross-validated comparison of the WoE logit with gradient boosting on the Taiwan data
- **Robustness**: other cut-offs, other measures (Brier, cost), group-wise AUC and calibration
- Example prompt
  - `Write Python code that compares a WoE logistic regression with gradient boosting on the Taiwan credit card default data by repeated stratified 5-fold cross-validation, recomputing the WoE bins inside each fold, and reports AUC, Brier score and the approval rates of men and women at a 75 percent approval cut-off.`

What to Check

- The coding of the outcome: default $= 1$; a reversed coding flips every sign and every curve
- The WoE sign convention: ln(goods/bads) here; some sources use ln(bads/goods)
- No leakage: bins, WoE and variable selection inside each training fold
- AUC is not accuracy, and Gini $= 2\,\text{AUC} - 1$, not AUC $- 0.5$
- PDs from an oversampled data set need the prior correction before any EL or capital number
- References: every cited paper and data set must exist; check the DOI

Project Idea

- **Question**: on the Taiwan credit card data, how much AUC does gradient boosting add over the WoE logit, and what happens to the approval rates by sex and age?
  - data: Default of Credit Card Clients (2009) (CC BY 4.0), saved in data/credit
- Steps
  - WoE bins and IV for every variable; a WoE logit; gradient boosting with default settings
  - repeated stratified cross-validation: AUC with DeLong intervals, Brier score, calibration
  - fairness at one cut-off: demographic parity and equal opportunity for sex and age groups
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected

Key Takeaways

- EL = PD × LGD × EAD is priced in; IRB capital covers the unexpected loss
- WoE turns every variable into log-odds evidence; IV ranks the variables
- Logit: e^{β} is an odds ratio; LDA (Fisher) gives the same linear log-odds under Normality and ranks equally well here
- A scorecard: Score = Offset + Factor ln(good:bad odds), PDO = 20 points per doubling
- Validate discrimination (AUC, Gini, KS) and calibration (Brier, reliability diagram) out of sample; choose the cut-off from costs
- Rejected applicants and protected groups need explicit checks; Altman and Merton add accounting and market information

Key Formulas

Quantity	Formula
EL	$\mathrm{PD} \times \mathrm{LGD} \times \mathrm{EAD}$
WoE, IV	$\ln \frac{g_i/G}{b_i/B}$, $\sum_i (g_i/G - b_i/B)\,\mathrm{WoE}_i$
Logit	$\ln \frac{p}{1-p} = \beta_0 + x^\top \beta$, $p = 1/(1+e^{-\eta})$
LDA	$w = S_W^{-1}(m_1 - m_0)$
Prior correction	$+\ln \frac{\pi/(1-\pi)}{\pi_s/(1-\pi_s)}$ to the constant
Scorecard	$\text{Offset} + \frac{\text{PDO}}{\ln 2} \ln \frac{1-p}{p}$
AUC, Gini	$P(s_{\text{bad}} > s_{\text{good}})$, $2\,\mathrm{AUC} - 1$
KS, Brier	$\max_s \lvert F_{\text{bad}}(s) - F_{\text{good}}(s)\rvert$, $\frac{1}{n}\sum (p_i - y_i)^2$
Bayes cut-off	$p > C_{FP}/(C_{FP} + C_{FN})$
Merton	$\mathrm{DD} = \frac{\ln(V/D) + (\mu - \sigma^2/2)T}{\sigma\sqrt{T}}$, $\mathrm{PD} = \Phi(-\mathrm{DD})$

Check Yourself

- **Question**: a model has an accuracy of 70% on the South German Credit test sample. Is it good?
  - **Answer**: not necessarily: "everybody is good" already reaches 70%; look at AUC, TPR, FPR and the cost
- **Question**: the coefficient of a dummy in a logit is 0.69. What does it mean?
  - **Answer**: the odds of default are multiplied by $e^{0.69} \approx 2$ for the group, relative to the reference group
- **Question**: does correcting the PDs for oversampling change the AUC?
  - **Answer**: no: it shifts all log-odds by the same constant, so the ranking is unchanged
- Next: Chapter 13, machine learning

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